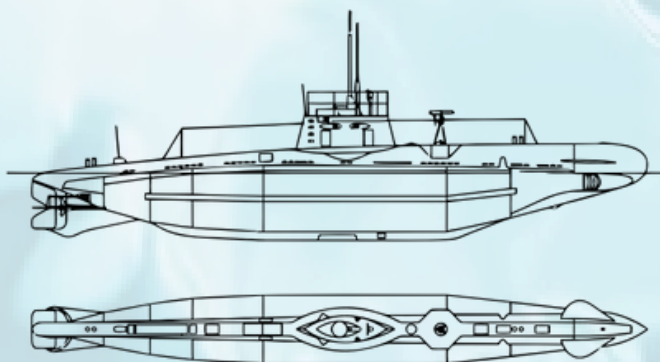




Sponsorship Package

UPR Mayagüez Campus
Autonomous Underwater Vehicle
2025 - 2026



University of Puerto Rico - Mayagüez
IEEE Student Branch



A Student Chapter of the IEEE Robotics & Automation Society



About Us

Founded in 2015 at the University of Puerto Rico Mayagüez Campus, RUMarino Autonomous Underwater Vehicle (RUMarino AUV) is a multidisciplinary team that specializes in designing, manufacturing and programming an autonomous underwater vehicle.

Team Objectives

Ignite and foster a passion for robotics and engineering.

Investigate real-world applications in Research & Development.

Offer chances to grow and take on leadership roles.

THREE-PILLAR MANDATE

SKILL ENHANCEMENT

- Introductory challenges for new team members
- Continuous mentorship on active projects
- Intern class members for skill development

COMMUNITY NETWORK

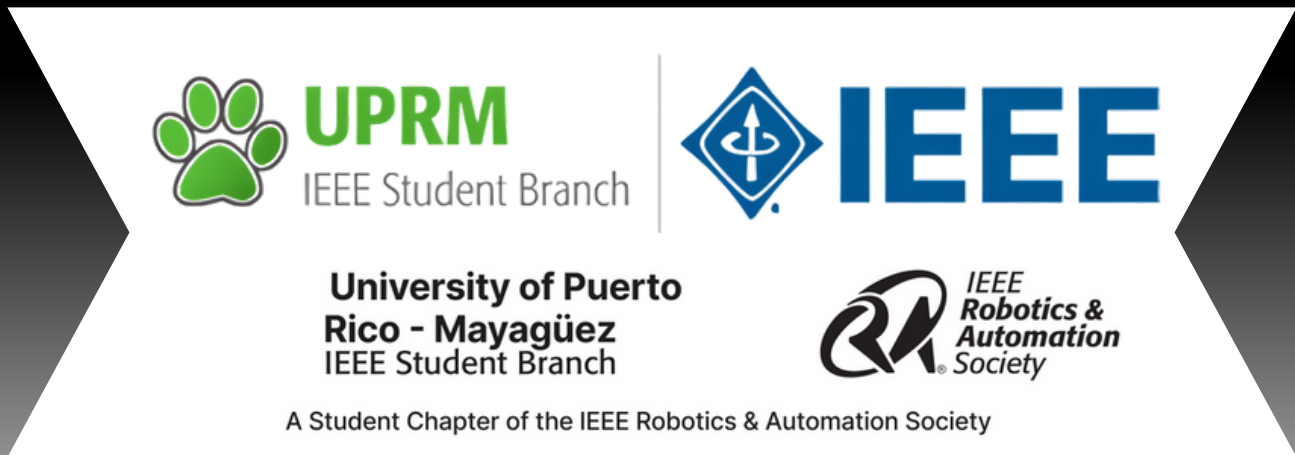
- Regular monthly and weekly social events
- Philanthropic efforts and educational outreach
- Cross-ESP and RoboSub teams

PROFESSIONAL & CAREER OPPORTUNITIES

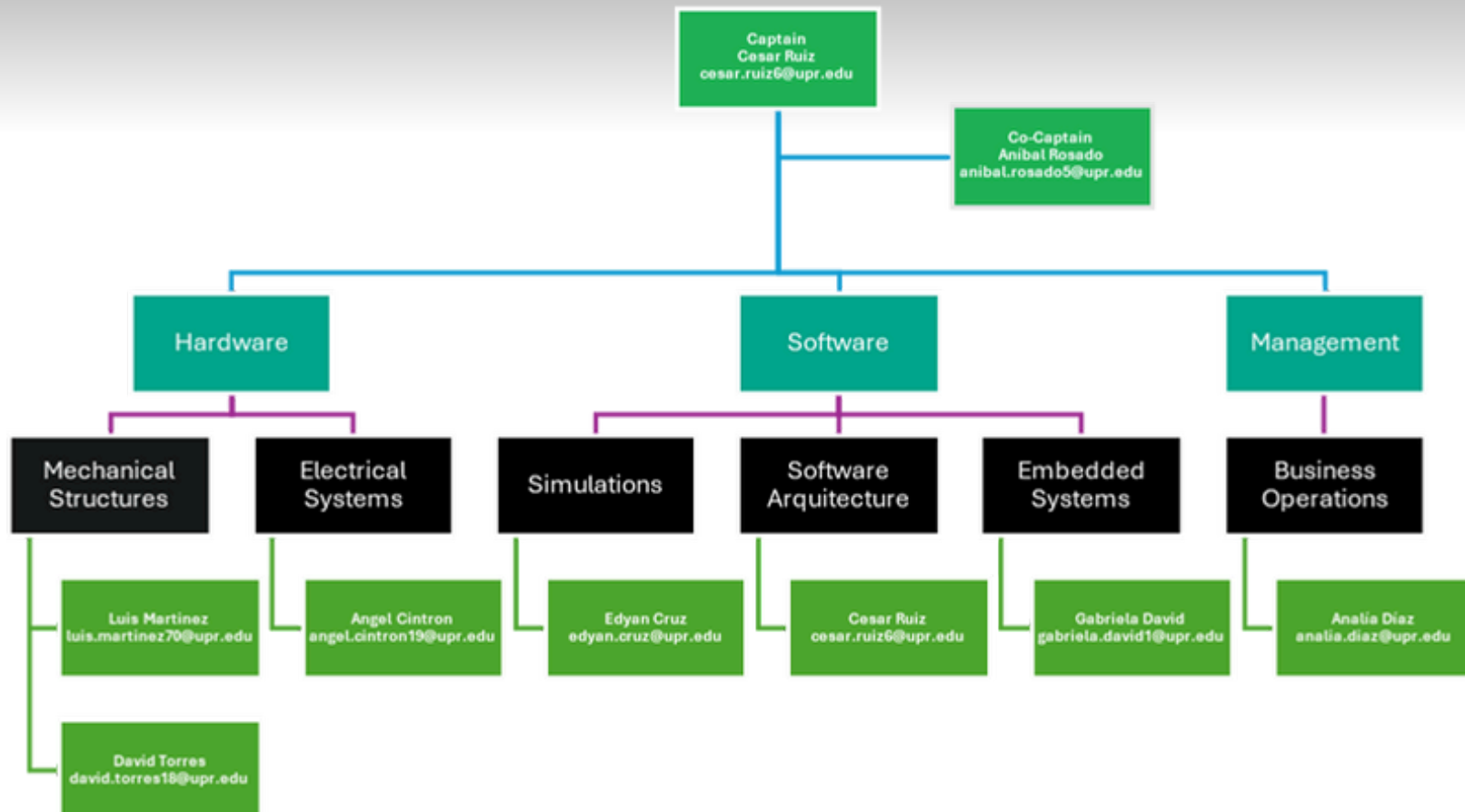
- Corporate & Sponsors site visits
- Partnerships with industry leaders
- Curated resume database



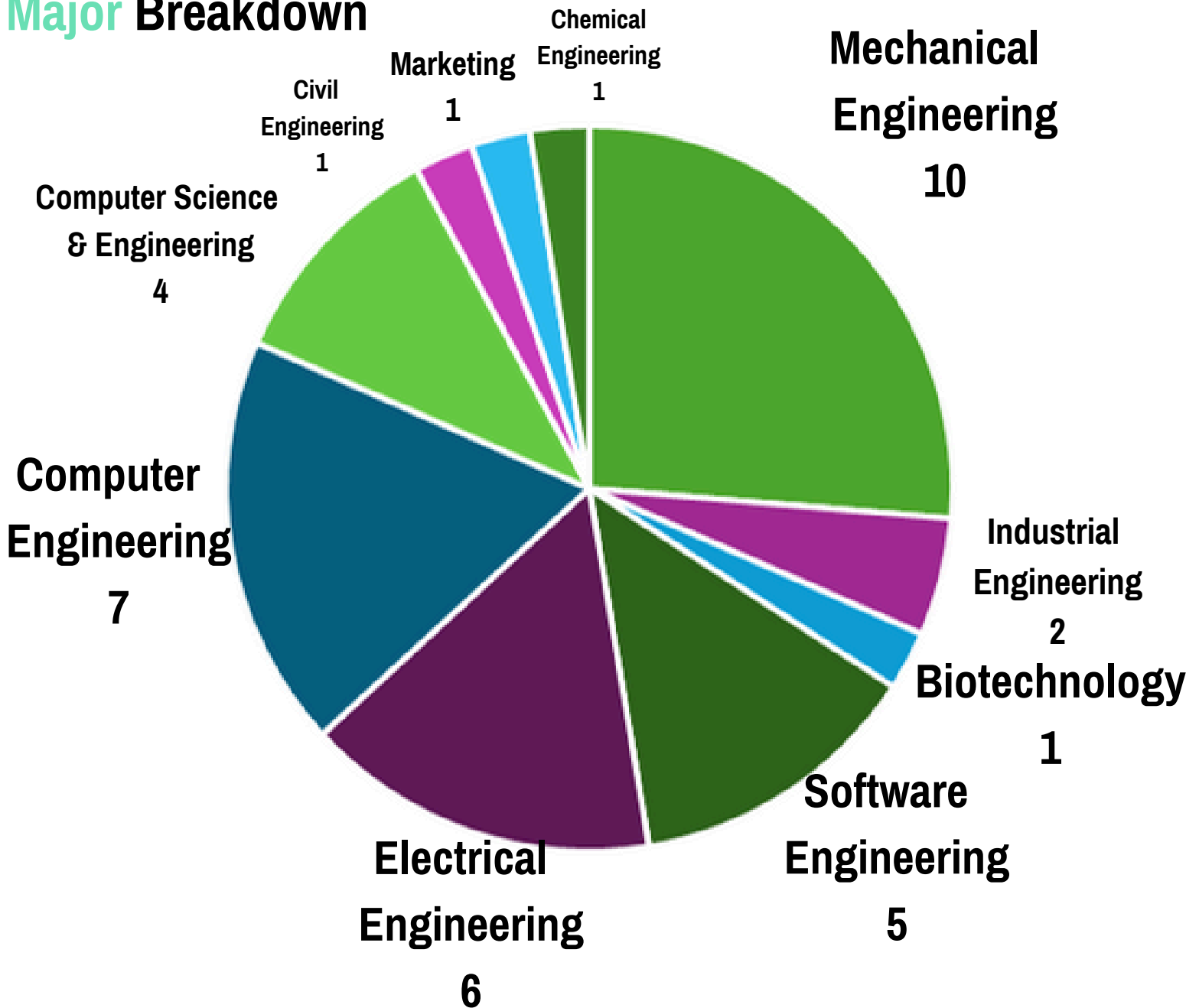
The goal when creating RUMarino was to build a student organization that helped engineering students use robotics and autonomous vehicles to put what they've learned in class into reality. To boost our goal RUMarino joined the IEEE and the IEEE Robotics & Automation Society.



Team Breakdown

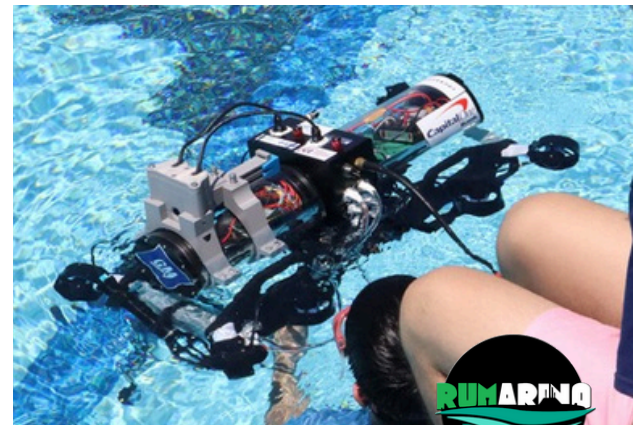


Major Breakdown

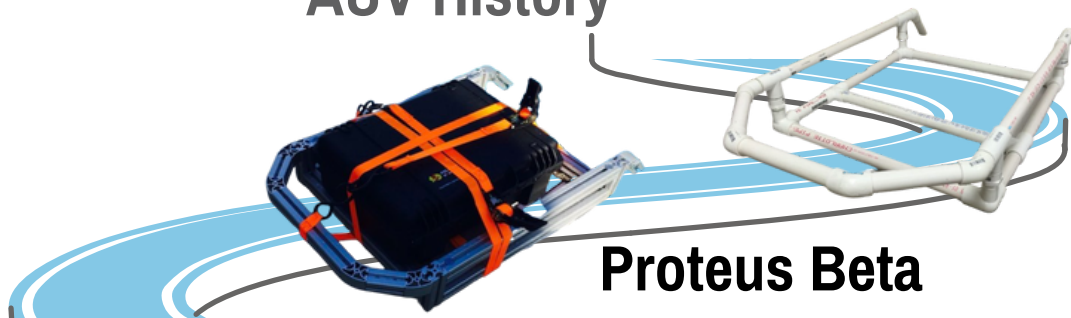


Member's School Year

1st 2nd 3rd 4th 5th 6th 7th



AUV History



Proteus Beta

Proteus 1.0

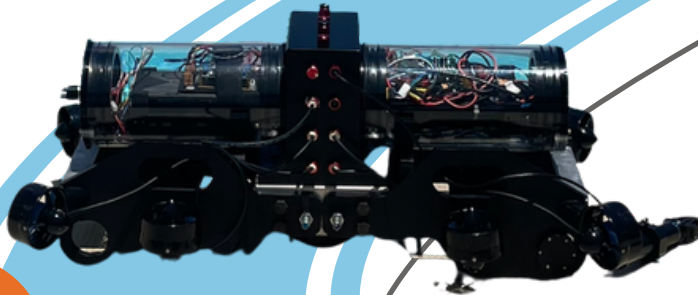


Proteus 2.0

Proteus 3.0

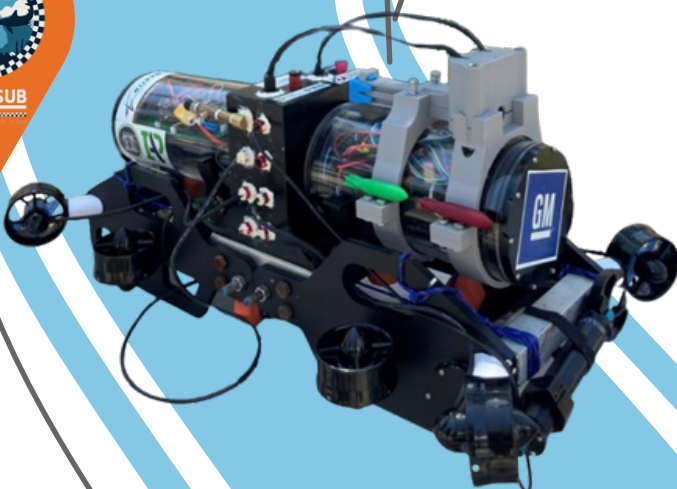


Hydrus Beta



Proteus

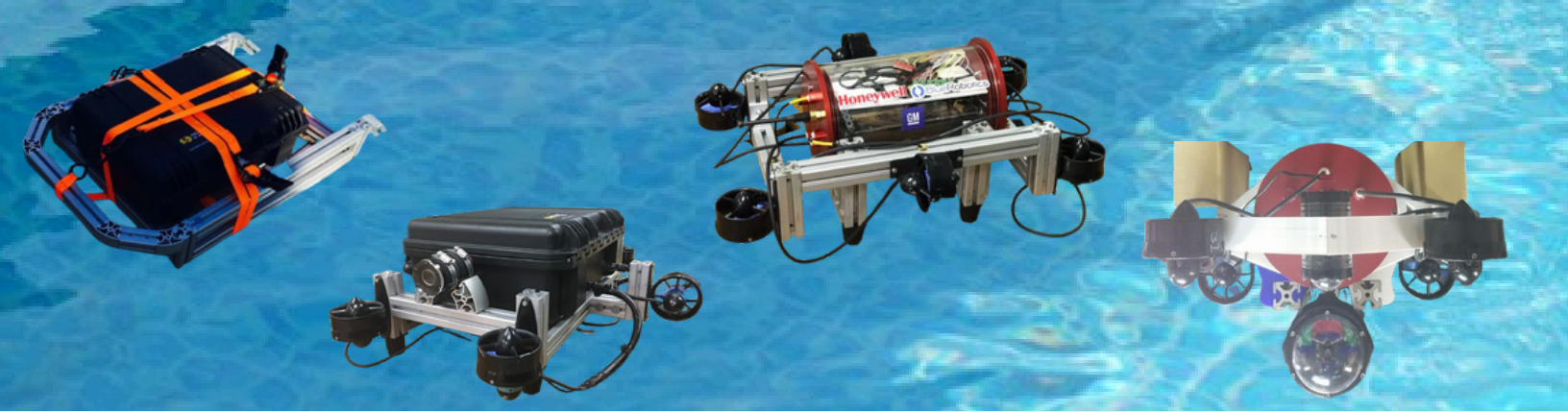
- 21st out of 44 competitors in 2017
- 17th out of 47 competitors in 2018
- 15th out of 59 participants in 2019



Hydrus

- 24th out of 41 participants in 2024
- 34th out of 58 participants in 2025





Proteus - AUV

Proteus is RUMarino's first AUV. Proteus has four main sensors which are: cameras used to detect the obstacles in a course; a pressure sensor to control the vehicle's depth; an Attitude and Heading Reference System (AHRS) used to control the heading movement of the vehicle; and finally, three hydrophones for acoustic signal detection of the underwater location of the source. In the competition, Proteus will have to complete different tasks and maneuver through obstacles. The tasks we face in the competition change every year. Proteus has successfully been tested autonomously and made quite a strong impression in its first appearance at the RoboSub 2017 competition, leading to awards in recognition of our performance. Currently, we are developing a new design for our new AUV with improved mechanical and electrical structures and a more robust software system.

Proteus 3.0 is our latest model for the 2018 RoboSub competition. Our new cabin for battery allocation, the modified mechanical structure, and our new software architecture allow for this AUV to be more robust, modular, and reliable.

Proteus 3.0 Components:

Arduino Due A000062

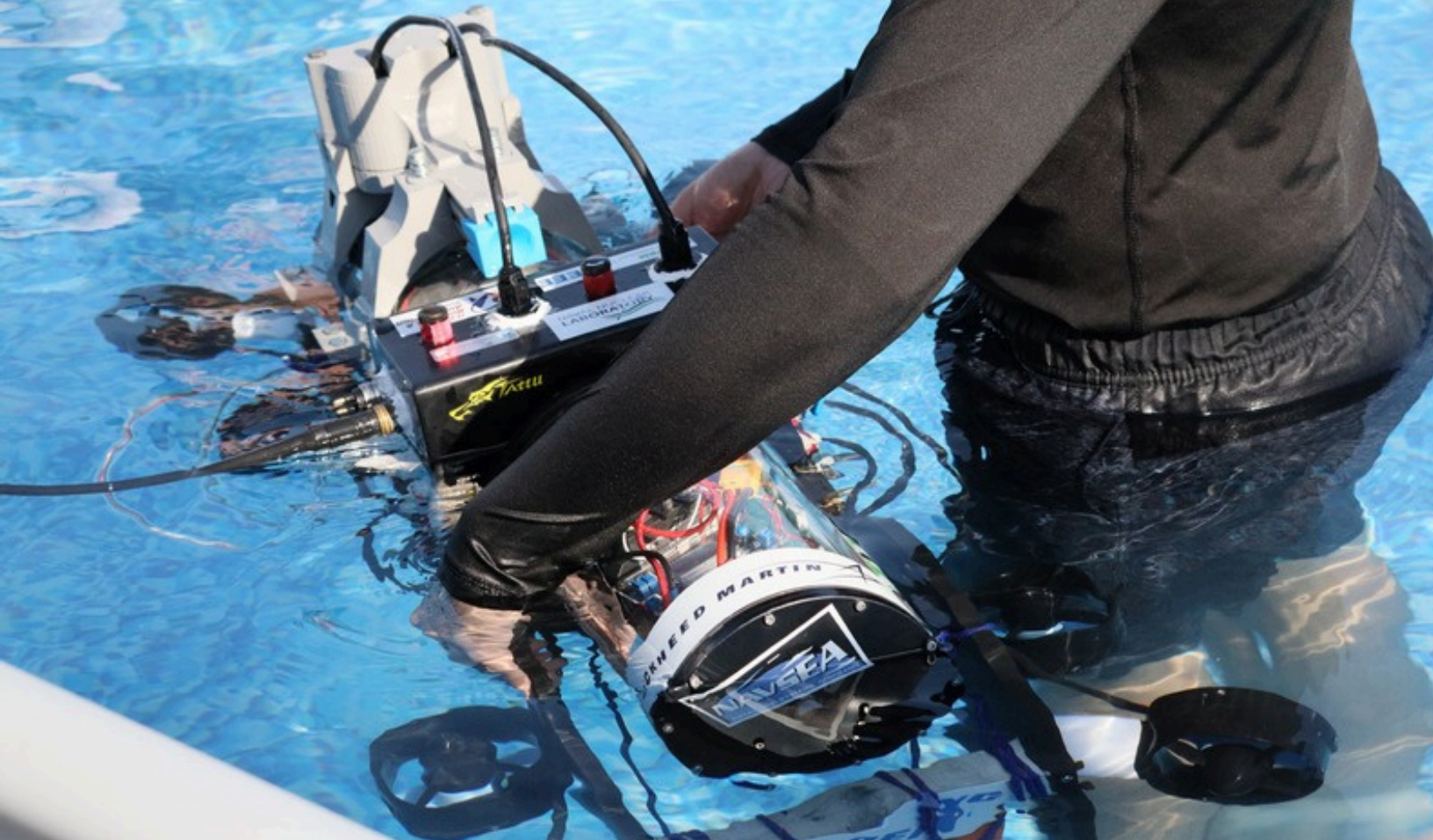
New water proof sub cabin

3D printed components

VectorNav VN-100 Rugged IMU

PCB circuitry





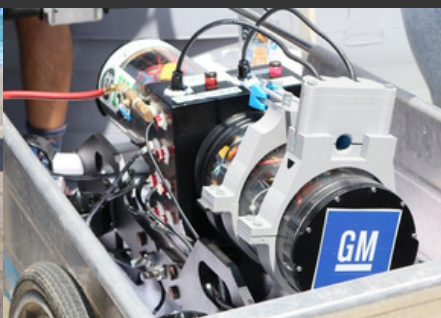
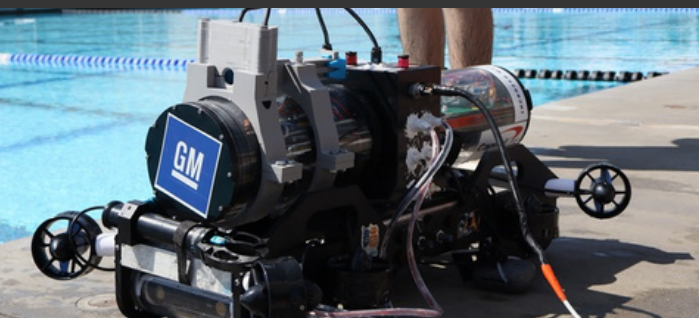
Hydrus - AUV

Hydrus is our newest Autonomous Underwater Vehicle designed with modularity and future expansion in mind. Measuring 3 feet long, carrying a stereoscopic camera, 8 thrusters, dual torpedo launchers, and a more extensive sensor suite, Hydrus is a complete leap forward in capability from its predecessor. It is the foundation for future improvements both in software and hardware and represents the cumulative progress that the team has achieved.

Hydrus has been in development since 2021 and for the first time it participated in RoboSub 2024. The team prepared Hydrus with the best autonomy to complete the tasks asked for the competition.

Components

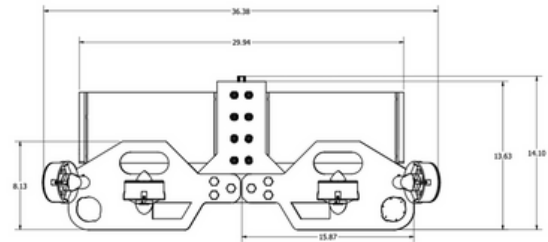
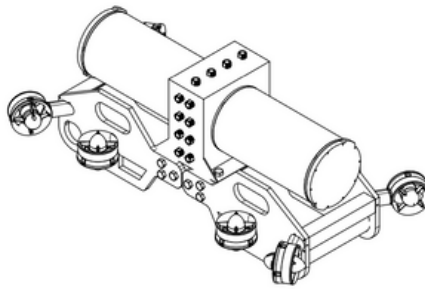
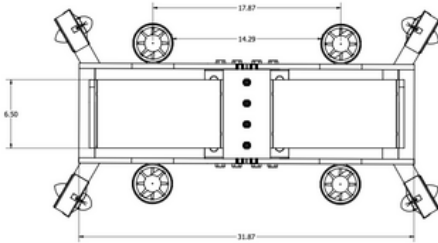
- Jetson TX2
- Zed 2i Camera
- Three new waterproof cabins
- PCB redesign
- Aluminum Structure
- Torpedo System
- Newton Subsea Gripper
- T100 motors



Next Gen. AUV

RUMarino is currently working on a new AUV design with the intention to have this submarine excel in the deficiencies that previous AUV's had.

We are aiming to include new components such as a new NVIDIA Jetson Nano, new T200 thrusters, new PCB boards and a improved power distribution system. New AUV designs aim to emphasize maneuverability, accessibility and modularity.



Timeline

Aug 2025

RoboSub
2025

Sep 2025

Next Gen AUV
Designs are
evaluated and
improved

Oct 2025

Internal
components are
established

Nov 2025

Designs are
reevaluated to
account for internal
components

Designs undergo
cost analysis
and reevaluation

Next Gen AUV
materials are
purchased

Dic 2025

Preparations
for RoboSub
2026 begin

Next Gen AUV is
manufactured
and tested

Spring 2026

Next Gen AUV is
evaluated for
competition
participation

RoboSub
2026

July 2026

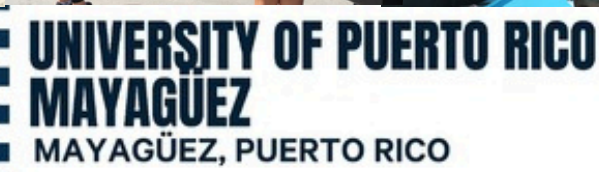




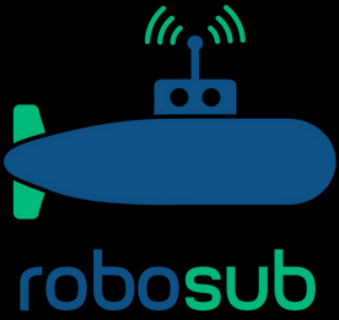
Robo-Sub Competition

Robo-Sub is an annual international competition where teams from all over the world gather at SSC Pacific's TRANSDEC facility in San Diego, California. While this event's main focus is finding the top AUV through a series of rigorous obstacle competitions, it also provides a platform for students to travel, network, and share ideas with others they'd normally not have a chance to. Students will put their AUV to the test by completing different missions. These range from precisely navigating through a gate, locating underwater signals, firing torpedoes, and many others. Once the missions begin, AUV's complete obstacles without any interaction from the participating students, and that's putting the "autonomous" in "autonomous underwater vehicle". The competition is co-sponsored by the Association for Unmanned Vehicle Systems International (AUVSI) and the U.S Office of Naval Research (ONR).

The performance of the AUV is not the only thing being evaluated by the judges, documentation presentation also comes in to play in the final standings of the competition. The teams are also evaluated by a technical design report, team website, a team video and a team presentation to the judges of the competition to give a summary of the technical design report while implying how the team will attack the obstacles of the competition.



Robo-Sub Achievements



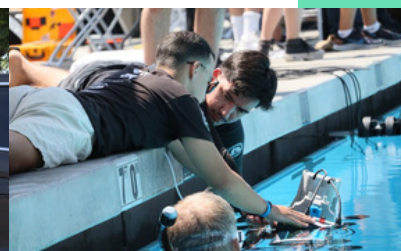
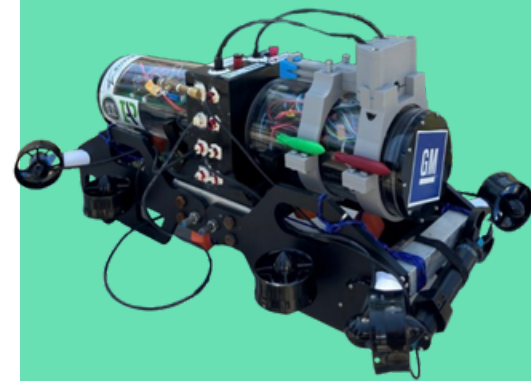
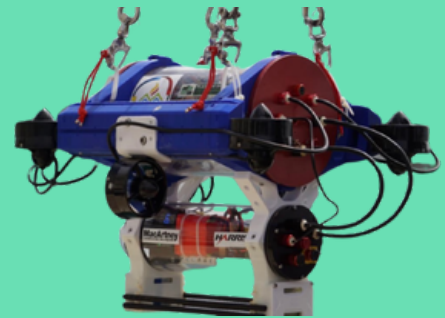
21st out of **44**
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competitors in 2018

15th out of **59**
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24th out of **41**
participants in 2024

34th out of **58**
participants in 2025



COMPUTER SYSTEMS



**Software
Architecture
Division**

**Simulations
Division**

**Embedded
Systems
Division**

SUB-TEAMS



**"Alone we can do so little; together we can do so much."
-Helen Keller**

HARDWARE



**Mechanical
Structures Division**

**Electrical
Systems Division**

MANAGEMENT



Business Operations

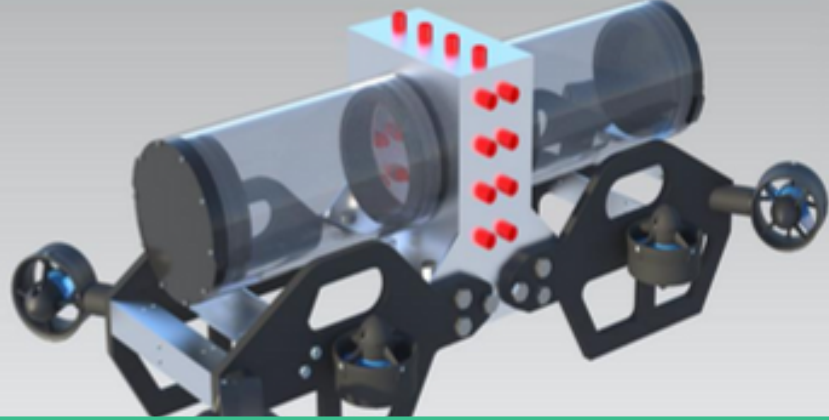


Computer Systems

Software is the brain of our AUV. From vision to motors, these are the spinal cords that combine everything together. The software collects data from sensors, processes it, and makes meaningful decisions from them. This team is composed by Computer Vision, Simulations, Software Architecture, Controls, Embedded Systems.

The Software Development Division handles the development of all the software within the AUV. This includes the mission logic and the vehicle's interactions with the environment. Some of the logic implemented was the mission controller, that is, a sequential program that decides what mission the vehicle will complete at a certain moment. Also, in this division most of the sensors such as the Doppler Velocity Logger (DVL), Inertial Measurement Unit (IMU) and ZED Stereo Vision Camera are implemented and modified for the software. Low level programming is also an important part of the team, as it works directly with the microcontrollers that make possible the movement of the vehicle. Currently, the team is designing and starting to implement a completely new approach to the tasks and obstacles for the 2026 RoboSub competition.

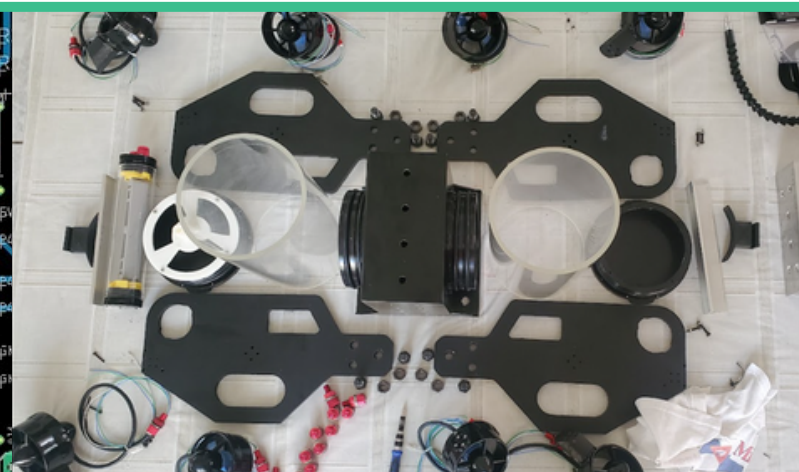
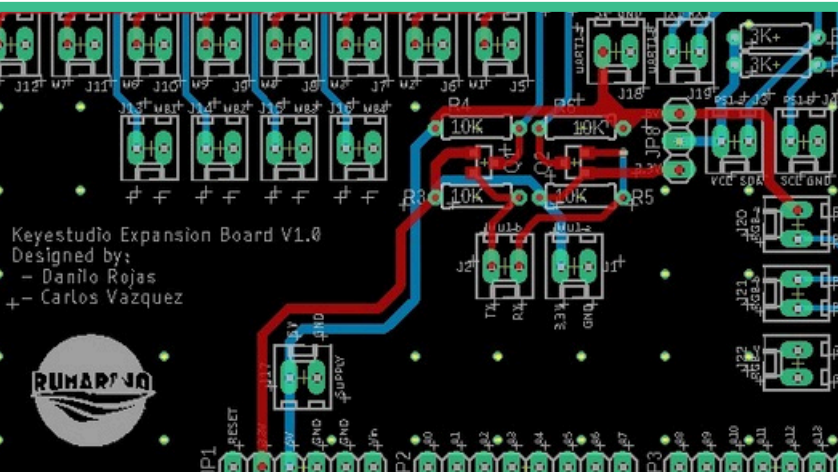




Hardware

The mechanical team is responsible for the mechanical integrity and functionality of the AUV. It ranges from the chassis to individual actuators such as thrusters, claws or torpedos. This division oversees the design of the whole vehicle in CAD software. It also does different analysis such as: Computer Fluids Dynamics (CFD) models, center of gravity calculations, Finite Element Analysis (FEA), and bouyancy calculations. The team also manufactures and builds the complete structure. This semester, the mechanical team has been completely redesigning the structure of our new AUV using Solidworks, Fusion 360, and other desing programs.

On the other hand, the electrical team is responsible for the organization of the interior layout of the electrical components, the design and implementation of the power distribution boards, and the design of the control system. Components such as: kill switches, voltages regulators, microcontrollers, and many others are implemented by this team. The team also designed the Printed Circuit Boards (PCB) in Proteus. The Electrical Systems Division is currently configuring new electrical systems in the Next Gen AUV. This new design will ensure a correct operation while fixing previous points of failure, making for a more consistent design.





Management

A division created to optimize the whole structure of the team. Its purpose is to mature RUMarino as it grows memberships, complexity and ambition. This team leads communication with potential sponsors, and makes relations between companies and the team. Operations Management optimizes the leading experience and procedures of the 9 multidisciplinary teams that compose RUMarino addressing their infrastructural, technological, personnel, and intellectual needs. This team also develops and oversees progress reports and process documentation for the future generations of RUMarino members to facilitate the transfer of knowledge from members that are graduating to new members in training by developing Standard Operating Procedures. Also, along with the Business Management Division, it relieves administrative responsibilities from the team leads and members that focus on developing the AUV.

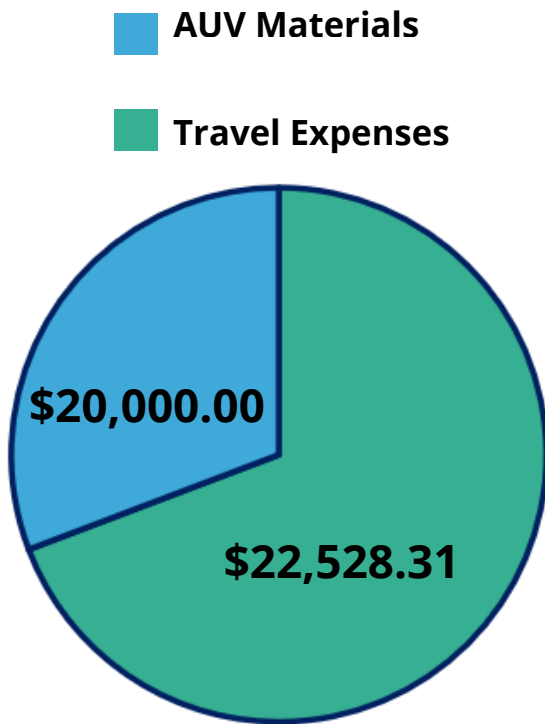
The business team manages the team resources such as revenue streams, documentation, and inventory which keeps the team operational and running efficiently. This team is also responsible of finding opportunities to promote Proteus, Hydrus and STEM fields for students in elementary, intermediate, and high school. This assures students to aspire to have further studies within these disciplines at the University of Puerto Rico, Mayagüez Campus. More information about the Team's Outreach activities can be found in our website: rumarino.org.



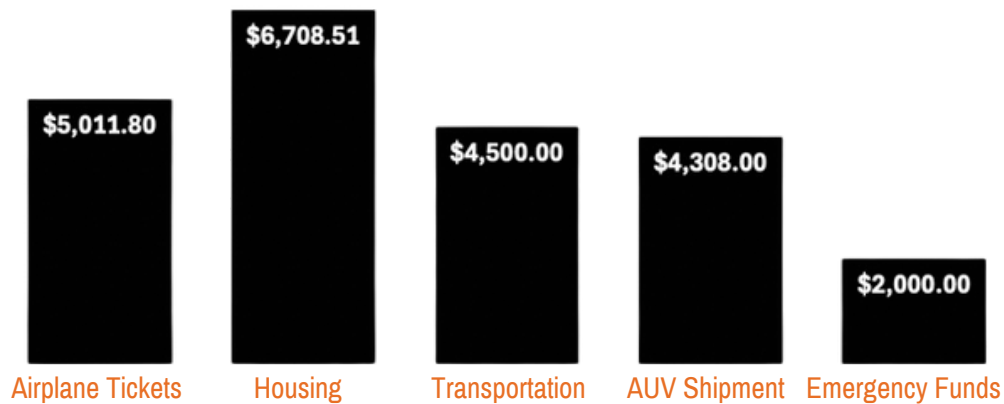
Cost and Expenses

Our general team's expenses are divided into AUV Materials and Travel Expenses, combined in a total of \$42,528.00. To help us cover our expenses, we encourage you to consider one of our four sponsorship tiers with exclusive benefits listed in the next page of this document.

General Team's Expenses



RoboSub Competition Expenses Glossary



Outreach

Campus Outreach

- Campus Project Fair
- Engineering Fun Day
- Pizza Sales

IEEE RAS Robotics Workshop

Sponsors Laboratory Visits

School Outreach Visits



**IEEE RAS & COMPUTER SOCIETY
ROBOTICS WORKSHOP**
"EXPLORING THE INTERSECTION OF TECHNOLOGY AND CREATIVITY"
FRIDAY FEBRUARY 16 - SUNDAY FEBRUARY 18
DATES AND ROOMS DETAILED BELOW:

DAY 1
FEBRUARY 16
4:30PM - 7:30PM
S-229

DAY 2
FEBRUARY 17
12:00PM - 3:00PM
S-211

DAY 3
FEBRUARY 18
12:00PM - 3:00PM
S-211

INSCRIPTION LINK CAN BE
FOUND IN OUR LINKTREE
PINNED IN OUR BOT

TALLER DE ANSYS
En este taller, exploraremos cómo Ansys optimiza diseños, analiza esfuerzos y mejora la eficiencia en proyectos de ingeniería. También se compartirán consejos sobre el uso de tecnología en el ámbito académico y profesional.

ESPACIOS LIMITADOS!!
Abierto a miembros de IEEE y proyectos asociados.

Cadlab Luchetti
Miércoles, 2 de abril de 2025
5:30pm

SCAN TO JOIN

Benefits of your Sponsor

1. Support the development of students in Science, Technology, Engineering and Mathematics (STEM), and Business Administration areas.
2. Promote the cutting-edge field of autonomous vehicles in Puerto Rico. The UPRM is the first university of Puerto Rico and Caribbean to compete in RoboSub.
3. Network with talented students with experience, knowledge and innovative ideas. Sponsors have already chosen members of RUMarino for internships and full time jobs in companies such as: Lockheed Martin, Boeing, Honeywell, GM, etc. due to their knowledge obtained within this project.
4. Sponsoring a nonprofit student organization in the University of Puerto Rico at Mayagüez is tax deductible to company contributions.
5. With us your company will be promoted even further. Team RUMarino has already grabbed the attention from multiple media outlets in Puerto Rico, for example, appearing in the front page of the local newspaper such as "ÍNDICE" and WAPA Televisions with Ada Monzón.

Sponsorship Tiers

List of Benefits	Bronze (\$50- 1,499)	Silver (\$1,500- \$2,499)	Gold (\$2,500- \$4,999)	Platinum (\$5,000+)
Company's logo will be on the AUV and on the team's shirt.		✓	✓	✓
Access to our student members resumes.	✓	✓	✓	✓
Framed picture at the end of the school year.			✓	✓
Access upon request to facilities in our campuses for meetings and events.				✓
Company's name on promotional displays and websites.	✓	✓	✓	✓
Company has the opportunity to drive the AUV.				✓

Other quantities of donations are accepted. List of benefits to accompany those are developed upon receipt.

Payment Methods

We appreciate all your support and sponsorship. The Payments method the Team accepts are the ones mentioned below. If any doubts, do not hesitate in contacting us.

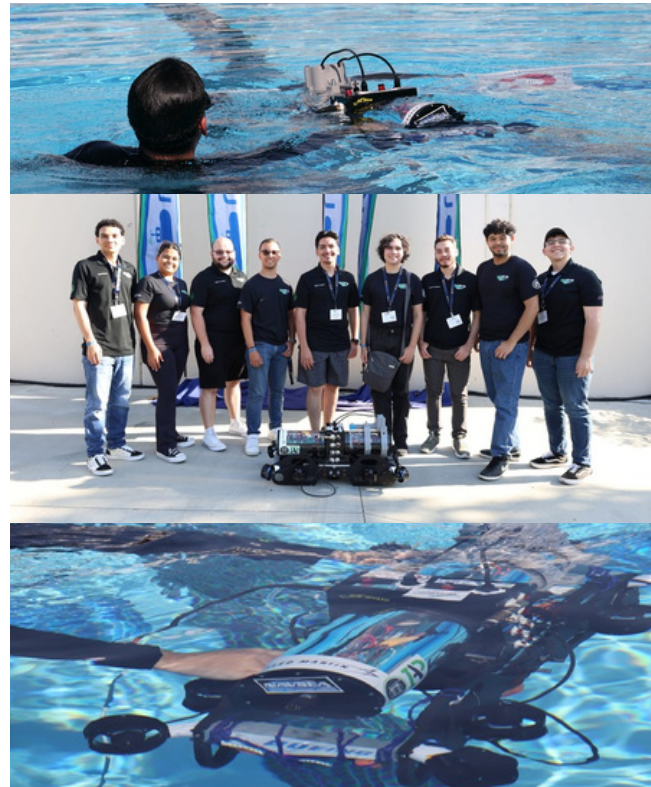
By Check

Make check payable to:

University of Puerto Rico - Mayaguez. Memo:
Rumarino Project - ECE Department

Send by Mail

Department of Electrical and Computer Engineering
University of Puerto Rico at Mayaguez,
Call Box 9000, Mayaguez, PR 00681



Sponsors



Primary Sponsor



Platinum (+\$5,000)



Gold (\$2,500-\$4,999)



Silver (\$1,500-\$2,499)



Bronze (\$50-\$1,499)

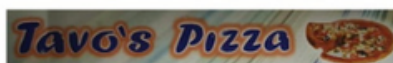
Other companies that have sponsored us:



Insurance • Finance • Investments™



L3HARRIS™
FAST. FORWARD.



Contact Us

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RUMarino Captain

Cesar A. Ruiz Navarro

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RUMarino Email

rumarino.uprm@gmail.com

RUMarino Co-Captain

Anbál E. Rosado Reyes

787-538-1604

anibal.rosado5@upr.edu



rumarino.org



rumarino_hydrus



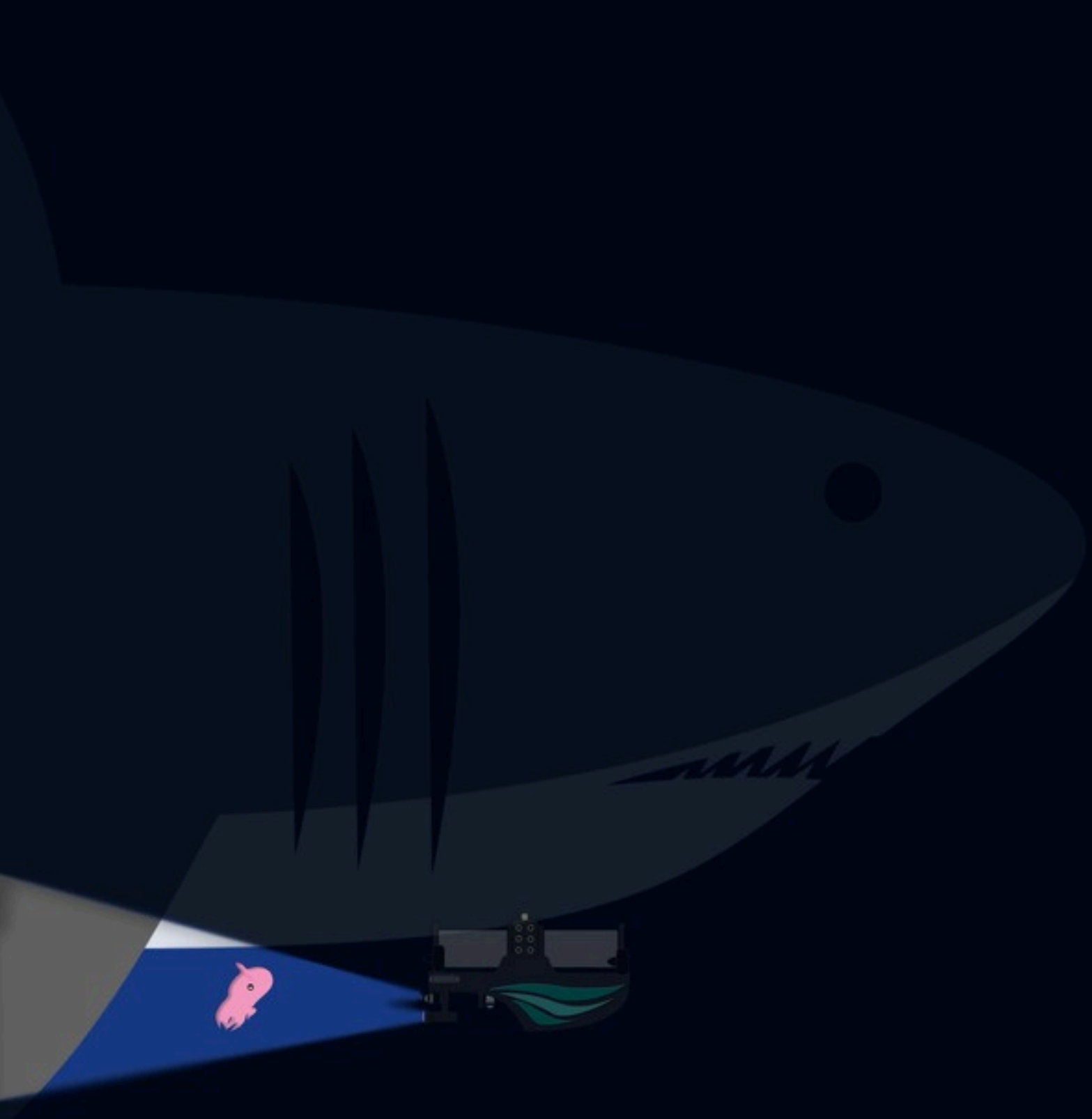
RUMarino



RUMarino_pr

Thank you for your time and support!





We own the Depths